Theory Of Cryptography

Written by HADIOUCHE Azouaou.

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This document contains the lectures given by Dr. OUDJIDA.

> To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

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Chapter 1

Introduction

Cryptography is the study of securing information so that the only parties that are authorized are able to access it. Multiple algorithms using difficult to solve problems from the theory of complexity to ensure data security in transmission. Here we will see the most known algorithms.

1.1. Secret & Public Key Cryptography

Imagine you and a friend would like to exchange objects on a daily basis, such that no one else can access it. So you think of two ideas:

- First, you will both have a single chest, and both of you have the key for it, and you can access it anytime you want while others are not allowed to given they do not have the key.
- Second, each of you will have their own chest and their own key that can be opened with it, while all the others have a key to close it. When you go to send an object to your friend, you put it in their chest and close it. No one except him can access the contents of the chest, and the same if he wants to send an object for you.

Both of those ideas are ways to do cryptography, the first being a secret key cryptography, that is, all parties have a secret key they use to encrypt and decrypt data with it. While the second is public key cryptography, each user in the network has a private / public key pair, other users use the public key to encrypt the data while the private key is used to decrypt it.

1.2. Mathematical Tools

Theorem 1.2.1 (Fermat's Little Theorem): *Let p be a prime number and a an integer such that $\gcd(a,p)=1$ then $a^{p-1} \equiv 1 \bmod p$.*

Notice that this can be used as a weak primality test, if a number q does not satisfy this identity, that is, $\exists a \in \mathbb{N}, \gcd(a,q)=1$ and $a^{q-1} \not\equiv 1 \bmod q$ then q is not prime. If $a^{q-1} \equiv 1 \bmod q$, we cannot conclude the primality of q, such numbers are called Carmichael numbers, and there exists infinitely many of them.

Definition 1.2.2 (Euler Totient Function): *The Euler totient function is the function $\Phi : \mathbb{N} \to \mathbb{N}$ defined as*

$$\Phi(n) = \#\{k \in [\![1,n]\!] \mid \gcd(k,n)=1\} = \#(\mathbb{Z}_n^*)$$

Lemma: *Let $r, m, p \in \mathbb{N}$ where p is prime, then $\gcd(m,p^r) \neq 1 \Leftrightarrow p \mid m$*

Proof. $\Rightarrow$ let $d = \gcd(m,p^r)$, since $d \mid p^r$ and p is prime then $d = p^k$ for some $k \in \mathbb{N}$, and $p^k = d \mid m$ thus $p \mid p^k$ and $p^k \mid m$ so $p \mid m$. $\Leftarrow$ suppose that $p \mid m$ and since $p \mid p^r$ too then $p \mid \gcd(m,p^r)$ given $p > 1$ then $\gcd(m,p^r) \neq 1$. □

Lemma: *Let $m, n, k \in \mathbb{Z}$, $\gcd(m,n)=1$ then*

$$\gcd(mn,k)=1 \Leftrightarrow \gcd(m,k)=1 \text{ and } \gcd(n,k)=1.$$

Proof. $\Rightarrow$ Suppose that $\gcd(mn,k)=1$, if $\gcd(m,k) \neq 1$ then there is a prime p such that $p \mid k$ and $p \mid m$, thus $p \mid mn$ too, so $p \mid k$ and $p \mid mn$, therefore, $p \mid \gcd(mn,k) \Rightarrow p = 1$, contradiction. $\Leftarrow$ Suppose that $\gcd(m,k)=1$ and $\gcd(n,k)=1$, by contradiction, assume that $\gcd(mn,k) \neq 1$ then $\exists p$ prime such that $p \mid mn$ and $p \mid k$, given that it is a prime then $p \mid m$ or $p \mid n$ which results in either $\gcd(m,k) \neq 1$ or $\gcd(n,k) \neq 1$, contradiction. □

Proposition 1.2.3:

1. *If p is prime, then $\varphi(p^r) = p^r\left(1 - \frac{1}{p}\right)$.*
2. *If a, b are coprime, $\varphi(ab) = \varphi(a)\varphi(b)$.*

Proof.

1. Let $p, r \in \mathbb{N}$ with p prime. From the lemma, we have that for $m \in \mathbb{N}$, $\gcd(p^r,m)=1 \Leftrightarrow p \nmid m$, the numbers that are divisible by p in $[\![0,p^r-1]\!]$

are of the form kp where $k \in [\![0, p^{r-1}-1]\!]$, so the number of numbers that are divisible by p in $[\![0, p^r - 1]\!]$ is $p^r - p^{r-1}$ so $\varphi(p^r) = p^r - p^{r-1}$.

2. $$\begin{aligned}\varphi(ab) &= \#\{k \in [\![1, ab]\!] \mid \gcd(k, ab) = 1\} \\ &= \#\{k \in [\![1, ab]\!] \mid \gcd(k, a) = 1 \text{ and } \gcd(k, b) = 1\} \\ &= \#(\{k \in [\![1, ab]\!] \mid \gcd(k, a) = 1\} \cap \{k \in [\![1, ab]\!] \mid \gcd(k, b) = 1\}) \\ &= \sum_{\substack{i=1 \\ \gcd(i,a)=1}}^{a} \sum_{\substack{j=1 \\ \gcd(j,b)=1}}^{b} 1 = \varphi(a) \cdot \varphi(b)\end{aligned}$$

□

Lemma 1.2.4: *Let $n \in \mathbb{N}$, $x \in (\mathbb{Z}/n\mathbb{Z})^*$ (invertible) $\Leftrightarrow \gcd(x, n) = 1$.*

Proof. $\Rightarrow$ Suppose that $x \in (\mathbb{Z}/n\mathbb{Z})^*$, then $\exists y \in \mathbb{Z}/n\mathbb{Z}$ such that $xy = 1 \bmod n$, then $xy = 1 + kn \Rightarrow xy - kn = 1$ by Bezout theorem we have $\gcd(x, n) = 1$. $\Leftarrow$ Suppose that $\gcd(x, n) = 1$, by Bezout theorem, $\exists y, k \in \mathbb{Z}$, $xy + kn = 1$ and thus $xy + kn = 1 \bmod n \Rightarrow xy = 1 \bmod n$ so $x \in (\mathbb{Z}/n\mathbb{Z})^*$. □

We need this lemma since $\varphi(n) = \#\,(\mathbb{Z}/n\mathbb{Z})^*$.

Theorem 1.2.5 (Euler): *Let m be a positive integer modulo a be an integer relatively prime to m then $a^{\varphi(m)} \equiv 1 \bmod m$.*

Proof. Let a relatively prime to m, then $\gcd(a, m) = 1 \Rightarrow a \in (\mathbb{Z}/m\mathbb{Z})^*$, a has finite order thus $\exists k \in \mathbb{N}, a^k = 1 \bmod m$, $\langle a \rangle$ is a subgroup of $(\mathbb{Z}/m\mathbb{Z})^*$ so $k \mid \#(\mathbb{Z}/m\mathbb{Z})^* = \varphi(m)$ and thus $\varphi(m) = kb$, thus $a^{\varphi(m)} = a^{kb} = \left(a^k\right)^b = 1^b = 1 \bmod m$, which completes the proof. □

Lemma 1.2.6: *Let $a_1, \ldots, a_k, x \in \mathbb{Z}$ such that $\forall i \neq j, \gcd(a_i, a_j) = 1$, and $\forall i \in [\![1, k]\!], a_i \mid x$ then $\prod_i a_i \mid x$.*

Proof. We prove it by induction on k.

- $k = 2$: Since $a_1 \mid x$ then $x = a_1 m$ for some $m \in \mathbb{Z}$, then we have that $a_2 \mid x \Rightarrow a_2 \mid a_1 m$, given that $\gcd(a_1, a_2) = 1$ then by Euclid's lemma we have that $a_2 \mid m$ so both $m = a_2 t$, replacing in x we get that $x = a_1 m = a_1(a_2 t) = (a_1 a_2)t$ thus $a_1 a_2 \mid x$.
- Suppose it is true for k: we have then a_{k+1} and $a_1 \cdots a_k$ are coprime, from the first case we have that $a_1 \cdots a_{k+1} \mid x$.

Therefore, we deduce that $\prod_i a_i \mid x$. □

Theorem 1.2.7 (Chinese Remainder Theorem): *Let $n_1, \ldots, n_k$ be pairwise coprime numbers and $N = n_1 \cdots n_k$ then*

$$\mathbb{Z}/N\mathbb{Z} \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$$

Proof. Consider the map $\varphi : \mathbb{Z} \to \prod_i \mathbb{Z}/n_i\mathbb{Z}, x \mapsto (x \bmod n_1, \ldots, x \bmod n_k)$, φ is clearly a well-defined ring homomorphism. Let $x \in \ker \varphi$ then $\forall i \in [\![1, k]\!], x \equiv 0 \bmod n_i$, from our lemma, we deduce that $N \mid x$, equivalently, if $N \mid x$, then $\forall i \in [\![1, k]\!], n_i \mid x$ so $\ker \varphi = N\mathbb{Z}$. To prove that φ is surjective, consider $(a_1, \ldots, a_k) \in \mathbb{Z}/n_1\mathbb{Z} \times \cdots \mathbb{Z}/n_k\mathbb{Z}$, define $N_i = N/n_i$, we have that $\gcd(N_i, n_i) = 1$ thus $\exists u_i, u_i N_i \equiv 1 \bmod n_i$, by taking $x = \sum_{i=1}^k a_i u_i N_i$ we obtain that $\varphi(x) = (a_1, \ldots, a_k)$ thus φ is surjective. Using the first isomorphism theorem, we get that $\mathbb{Z}/N\mathbb{Z} \cong \prod_i \mathbb{Z}/n_i\mathbb{Z}$. □

Corollary 1.2.8 (CRT / System Of Congruences): *Let $n_1, \ldots, n_k$ pairwise coprime integers, $N = n_1 \cdots n_k$, and $a_1, \ldots, a_k$ be integers, then*

$$\begin{cases} x = a_1 \bmod n_1 \\ x = a_2 \bmod n_2 \\ \quad \vdots \\ x = a_k \bmod n_k \end{cases}$$

has a unique solution for x as follows

$$N_i = N/n_i \qquad u_i N_i \equiv 1 \bmod n_i$$

$$x = \sum_{i=1}^{k} a_i u_i N_i \bmod N$$

Chapter 2

RSA Cryptosystem

Here we introduce the most known cryptosystem, the RSA, which is an algorithm that depends on the difficulty of factorizing a number into its divisors.

2.1. Mathematical Formulation Of RSA

Theorem 2.1.1: *Let $p, q \in \mathbb{N}$ be two primes, $n = pq$, $e \in \mathbb{N}$ such that $\gcd(e, \varphi(n)) = 1$ and $d \in \mathbb{N}, ed \equiv 1 \bmod \varphi(n)$. Then for any $m \in \mathbb{N}$ we have that if $c \equiv m^e \bmod n$ then $m \equiv c^d \bmod n$.*

Proof. Let $c \equiv m^e \bmod n$, we have first that $ed \equiv 1 \bmod \varphi(n)$ thus $ed = \varphi(n)k + 1$, then $c^d \equiv (m^e)^d \equiv m^{ed} \equiv m^{\varphi(n)k+1} \equiv \left(m^{\varphi(n)}\right)^k m \equiv m \bmod n$, in the last equality we used Euler's theorem to have that $m^{\varphi(n)} \equiv 1 \bmod n$. □

Notice that we obtain the class of m after the decryption, so to make this mapping unique, we consider the message space to be $[\![0, n-1]\!]$ thus the mapping $f(m) = m^e \bmod n$ becomes a bijection.

2.2. Algorithmic Implementation Of RSA

Algorithm:

- ***Key Generation:***
  1. *Generate p, q prime numbers.*
  2. *Calculate $n = pq$.*
  3. *Calculate $\varphi(n) = (p-1)(q-1)$.*
  4. *Generate e such that $\gcd(e, \varphi(n)) = 1$.*
  5. *Calculate d such that $ed \equiv 1 \bmod \varphi(n)$.*
  6. *Public Key: e, n, Private Key: p, q, d.*

- ***Encryption:***
  1. *Let $m \in \mathbb{N}$ be a message.*
  2. *If $m > n$, divide it into parts that are.*
  3. *Encrypt $\mathrm{RSA}_{e,n} \equiv m^e \bmod n$.*

- ***Decryption:***
  1. *Let $c \in \mathbb{N}$ an encrypted message.*
  2. *Decrypt $\mathrm{RSA}^{-1}_{d,n} \equiv m^d \bmod n$.*

The correctness of this algorithm comes from the previous theorem and the fact that we consider only message of length at most $n-1$ to have uniqueness.

2.2.1. Optimization Methods

Notice that in both encryption and decryption, we use modular exponentiation, that is, we calculate $a^b \bmod n$ for some $a, b, n \in \mathbb{N}$. We have the following proposition that makes this calculation easier.

Proposition: *Let $a, b, n \in \mathbb{N}$, and let $b = b_1 + \cdots + b_k$, then*

$$a^b \equiv \left(\prod_{i=1}^{k} (a^{b_k} \bmod n)\right) \bmod n$$

Proof. We start by proving that $ab \equiv (a \bmod n)(b \bmod n) \bmod n$, indeed, we have that $a = nq_a + r_a$ and $b = nq_b + r_b$ with $0 \le r_a < q_a, 0 \le r_b < q_b$ by the Euclidean division, thus $ab = (nq_a + r_a)(nq_b + r_b) = n^2 q_a q_b + n(q_a r_b + q_b r_a) + r_a r_b$ and notice that $r_a = a \bmod n$ and $r_b = b \bmod n$ thus $ab \equiv r_a r_b = (a \bmod n)(b \bmod n) \bmod n$. By induction we have that $\left(\prod_i a_i\right) \bmod n = \left(\prod_i (a_i \bmod n)\right) \bmod n$.

We have $a^b = a^{b_1 + \cdots + b_k} = \prod_{i=1}^{k} a^{b_i}$, and by the previous point, we have that $a^b \equiv \left(\prod_{i=1}^{k} a^{b_i}\right) \equiv \left(\prod_{i=1}^{k} (a^{b_i} \bmod n)\right) \bmod n$. □

Notice that using this, if we consider an improved decomposition of b, we can get an optimization in the calculation of the modular exponentiation over the naive approach of just repeated multiplication $\bmod n$. We consider in this case the binary decomposition $b = \sum_{i=0}^{k} b_i 2^i$ with $b_i \in \{0, 1\}$. Thus, we get

$$a^b = a^{\sum_{i=0}^{k} b_i 2^i} = \prod_{i=0}^{k} a^{b_i 2^i} = \prod_{i=0}^{k} \left(a^{2^i}\right)^{b_i}$$

and so we get

$$a^b \equiv \left(\prod_{i=0}^{k} \left(a^{2^i}\right)^{b_i} \bmod n\right) \bmod n$$

so the only factors that matter is when $b_i = 1$, and we have $a^{2^{i+1}} = \left(a^{2^i}\right)^2$. Hence, we get the following algorithm for calculating the modular exponential.

Algorithm (Square-And-Multiply Algorithm):

```
input:
  - base: a
  - exponent: b
  - modulus: n
output:
  - c = a^b mod n

algorithm:
  res = 1
  base = a % n

  while b > 0:
    if b % 2 = 1:
      res = (res * base) mod n

    base = (base * base) mod n
    b = b / 2

  return res
```

As discussed before, base here refers to a^{2^i} in each iteration it is squared, and we add it as a factor of the product of the result if and only if $b \equiv 1 \bmod 2$ which is equivalent to saying $b_i = 1$. Now to prove the efficiency, let's calculate the complexity of each methods.

Proposition: *The complexity of the algorithms is as follows:*

- *Naive Exponentiation:* $O(b)$.
- *Binary Exponentiation (Square-And-Multiply):* $O(\log(b))$.

Proof. In the calculation of complexity, we consider that the multiplication of two arbitrary numbers is $O(1)$ which is a simplification.

- We multiply a by itself $b-1$ times, thus the loop has $O(b)$ steps.
- When we turn a number b into its binary representation, we have $\log_2(b)$ digits and given that we do $\log_2(b)$ iterations in the algorithm, and each iteration has at most 3 operations then it is of order $O(\log_2 b)$.

□

This is asymptotically optimal, since any algorithm would need to read $\log_2(b)$ digits at least thus binary exponentiation is optimized.

Another optimization that we can do is in the decryption which is a computationally expansive procedure given that it will be a huge number to the power of another huge number, thus we use the Chinese Remainder Theorem to simplify the calculations by a divide-and-conquer algorithm.

Proposition: *With the same context as Theorem 2.1.1, let* $m_p \equiv c^d \bmod p$ *and* $m_q \equiv c^d \bmod q$*, let* $q' \in \mathbb{Z}$ *such that* $q'q \equiv 1 \bmod p$ *then we have*

$$m \equiv \left(m_q + q \cdot \left(q'\left(m_p - m_q\right) \bmod p\right)\right) \bmod n$$

Proof. We use the CRT to find m that satisfies $m \equiv m_p \bmod p$ and $m \equiv m_q \bmod q$. We start from the first equation that $m = m_q + q \cdot k$, by replacing in the second equation we have that $m_q + q \cdot k \equiv m_p \bmod p \Rightarrow q \cdot k \equiv m_p - m_q \bmod p \Rightarrow k \equiv q'(qk) \equiv q'\left(m_p - m_q\right) \bmod p$, by replacing in the second equation $m = m_q + q \cdot \left(q'\left(m_p - m_q\right) \bmod p\right)$ and thus we get the class of m being $m_q + q \cdot \left(q'\left(m_p - m_q\right) \bmod p\right) \bmod n$. □